



Basic Principles of Medicine 1
Module: Musculoskeletal System | Lecture 9

Imaging of the Thorax

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Objectives

- SOM.MK.2.BPM1.1.MSK.1.ANAT.1
235** Identify the vertebrae, ribs, sternum, clavicle, anterior and posterior thoracic wall, on imaging studies.
- SOM.MK.2.BPM1.1.MSK.1.ANAT.1
236** Identify the muscles of the thoracic wall on imaging studies.
- SOM.MK.2.BPM1.1.MSK.1.ANAT.1
237** Identify rib fractures on imaging studies.
- SOM.MK.2.BPM1.1.MSK.1.ANAT.1
238** Describe Flail chest, its mechanism of action and use this information to interpret clinical(radiographic) signs and symptoms.



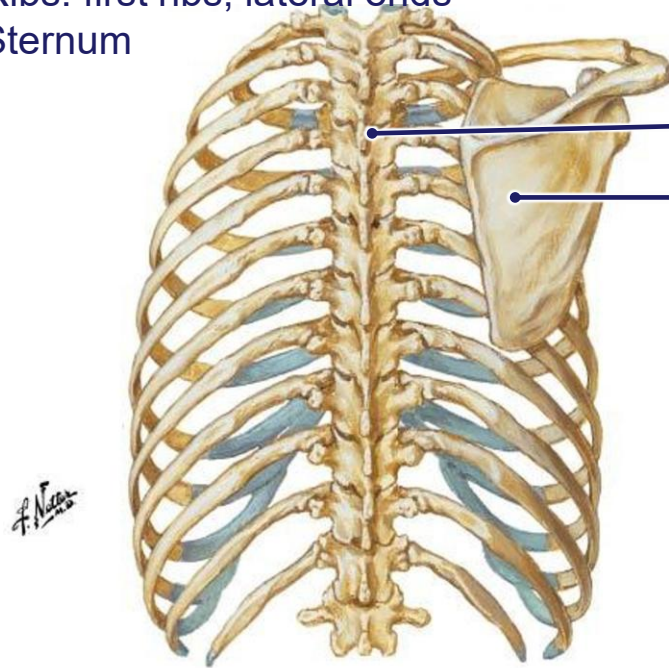
Recommended Reading

- Drake, Vogl, Mitchell. Gray's Anatomy for students. 4th Ed. Elsevier.
- Printed Textbook: Chapter 3 pg 140-168
- Online Textbook: Chapter 3
 - Conceptual Overview
 - Regional Anatomy
 - Pectoral region
 - Thoracic wall
 - Diaphragm
 - Movements of the thoracic wall and diaphragm during breathing
 - Pleural cavities
- Pay special attention to the green “In the Clinic” boxes for clinical correlations related to this topic

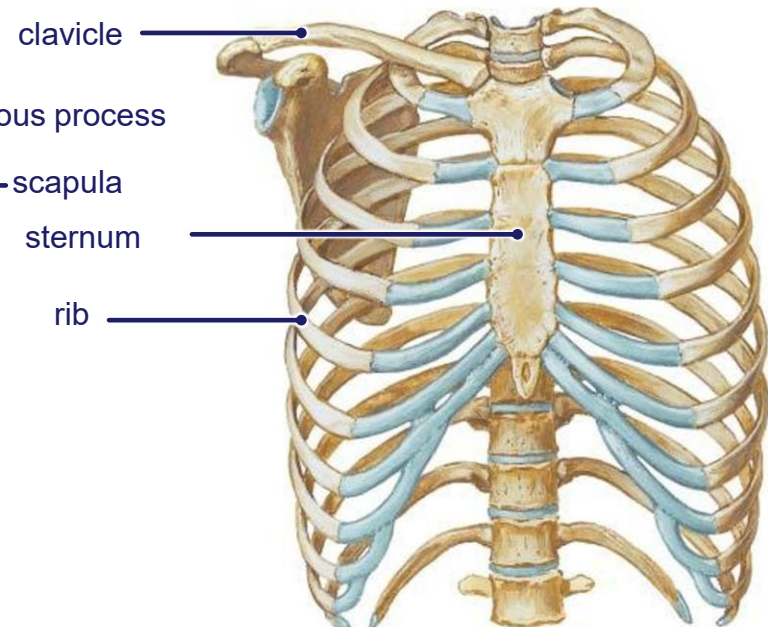
Bony landmarks

Landmarks

- Scapula
- Clavicle
- Vertebral spinous processes
- Ribs: first ribs, lateral ends
- Sternum



Posterior view



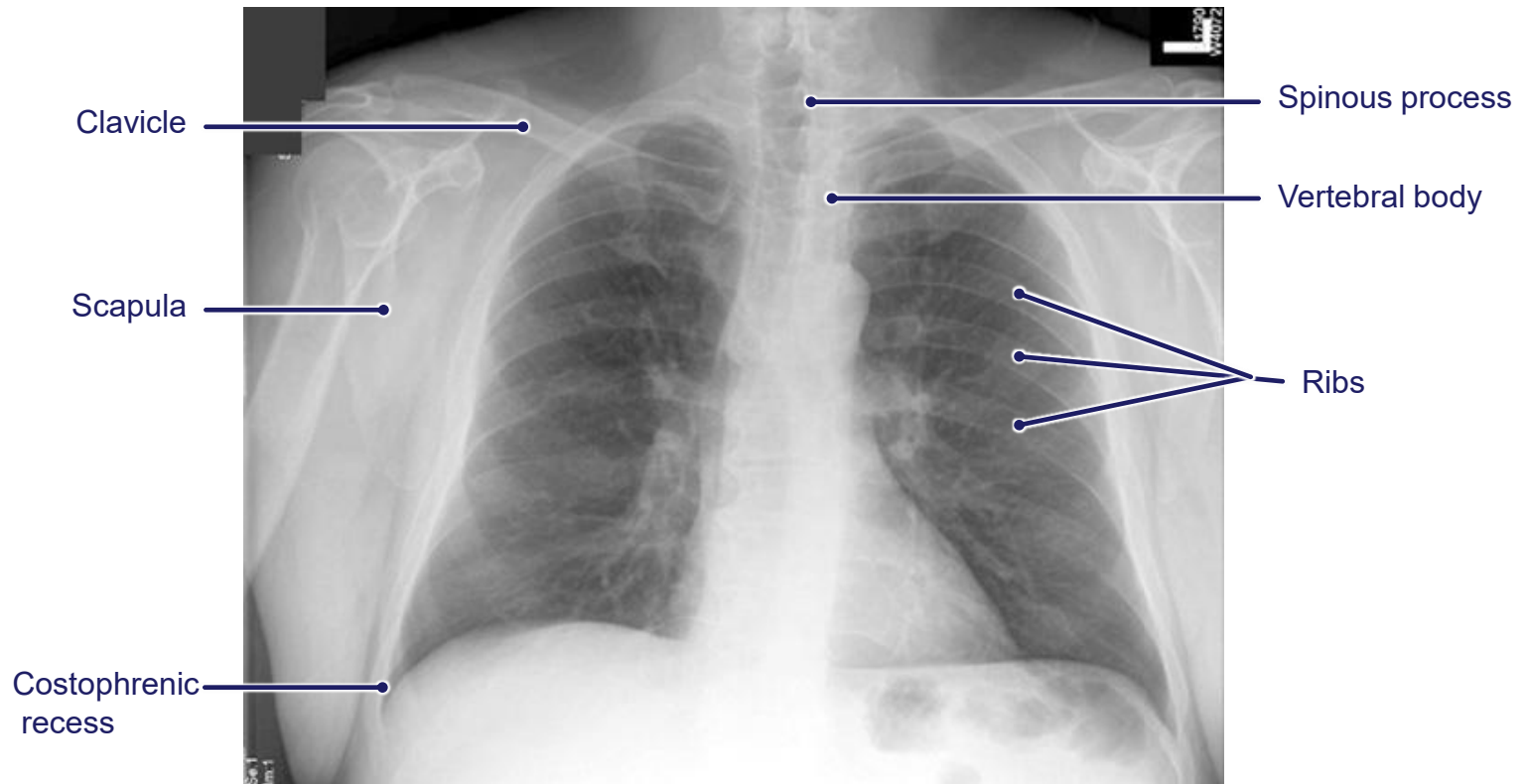
Anterior view

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Chest X-ray

Landmarks

- Scapula
- Vertebral spinous processes midway between the medial ends of the clavicle
- Ribs: first ribs, lateral ends
- Costophrenic recesses
- Sternum

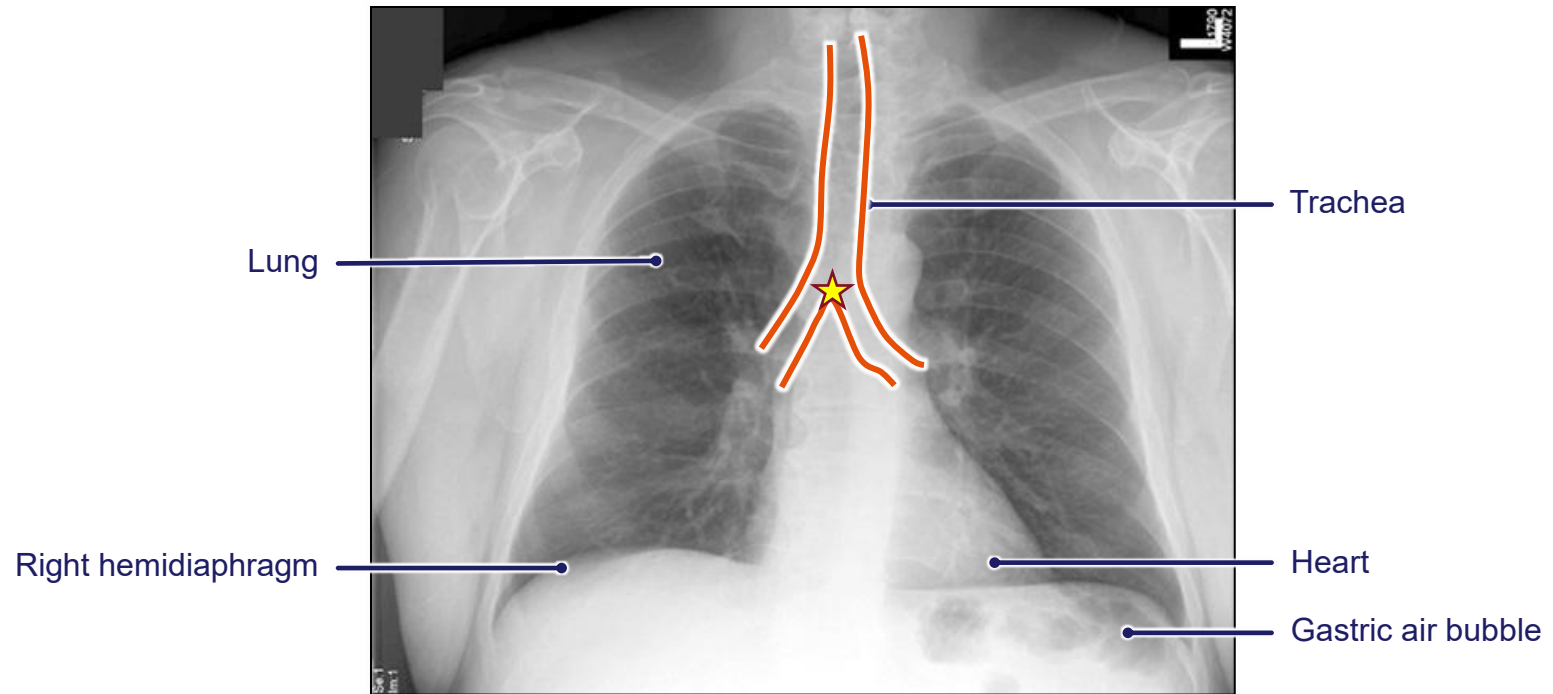


Posterior-Anterior (PA) view

Chest X-ray

Landmarks

- Lungs
- Hilum
- Trachea
- Heart
- Diaphragm
- Gastric air bubble



Posterior-Anterior (PA) view

Differentiating between different aspects of the ribs

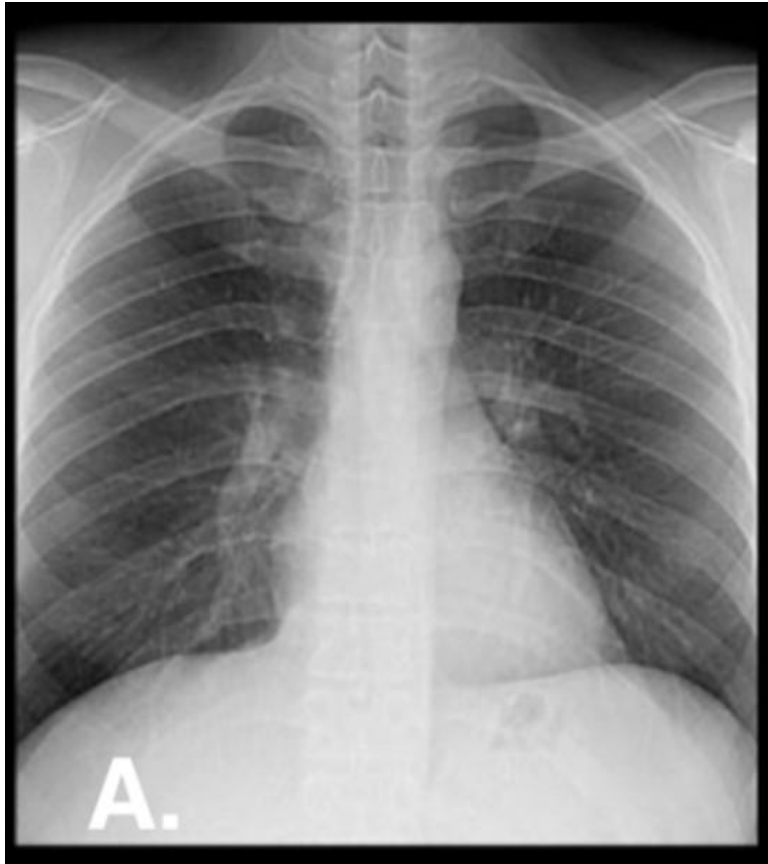
ANTERIOR

- Visible but are more difficult to see
- Oriented downward (obliquely)
- Attach to sternum or each other with **cartilage** that is usually not visible on x-rays
 - May calcify later in life and will then be visible

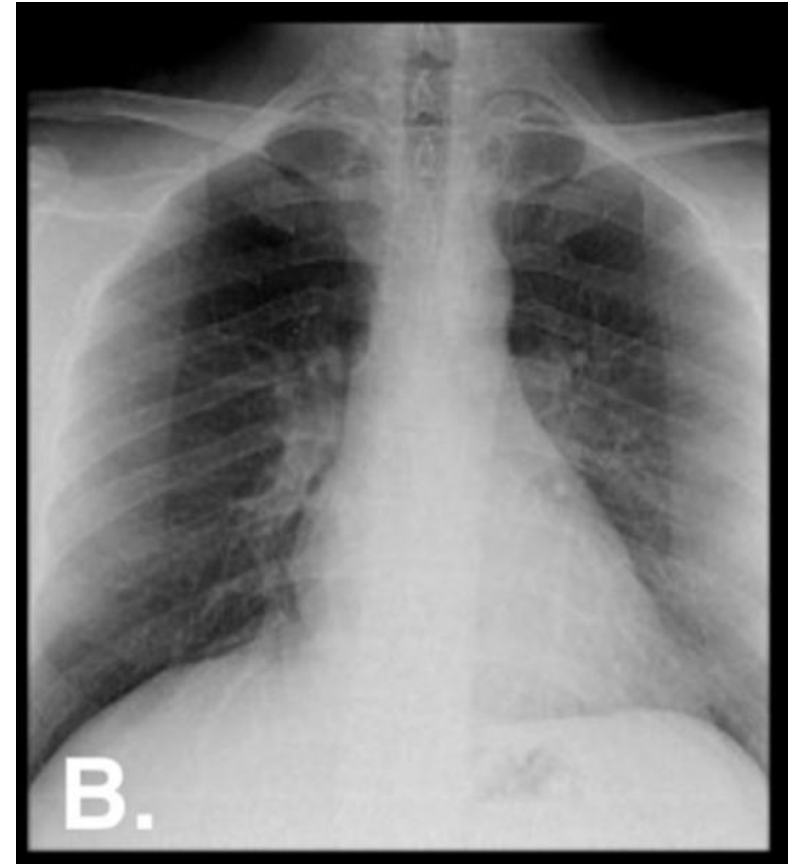
POSTERIOR

- Immediately more apparent on frontal chest radiographs
 - Usually the ones that are counted
- Oriented more or less horizontally
- Each pair of posterior ribs attaches to a thoracic vertebral body

Projections



Postero-anterior (PA): central ray passes from posterior to anterior



Antero-posterior (AP): central ray passes from anterior to posterior

<https://radrounds.com/radiology-case-images-teaching-file/chest-x-ray-basics-pa-vs-ap/>

Chest X-ray Artefacts

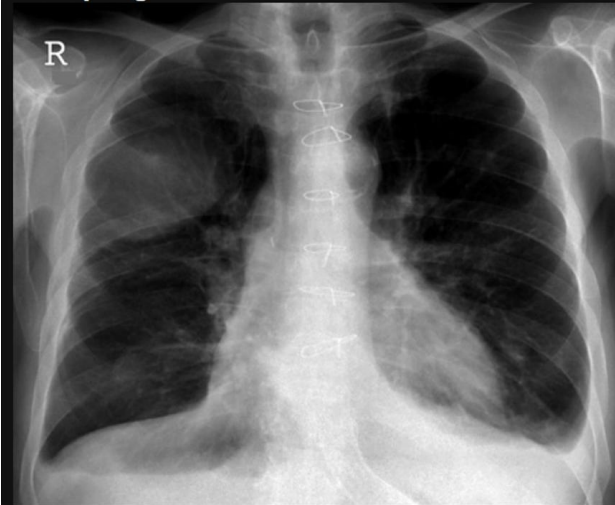
Normal reference



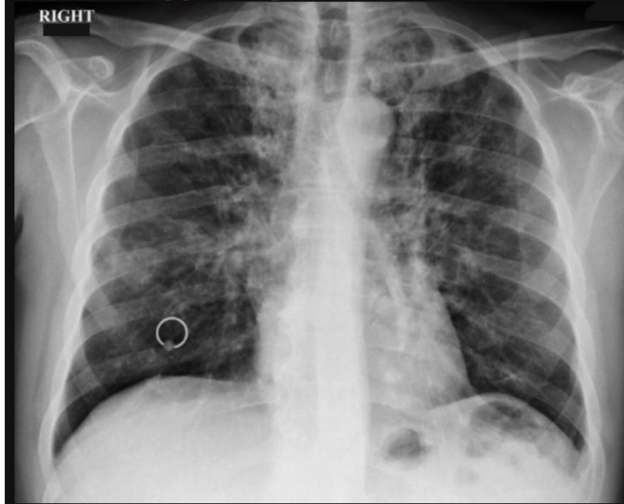
Jugular line for chemotherapy



Oesophageal stent

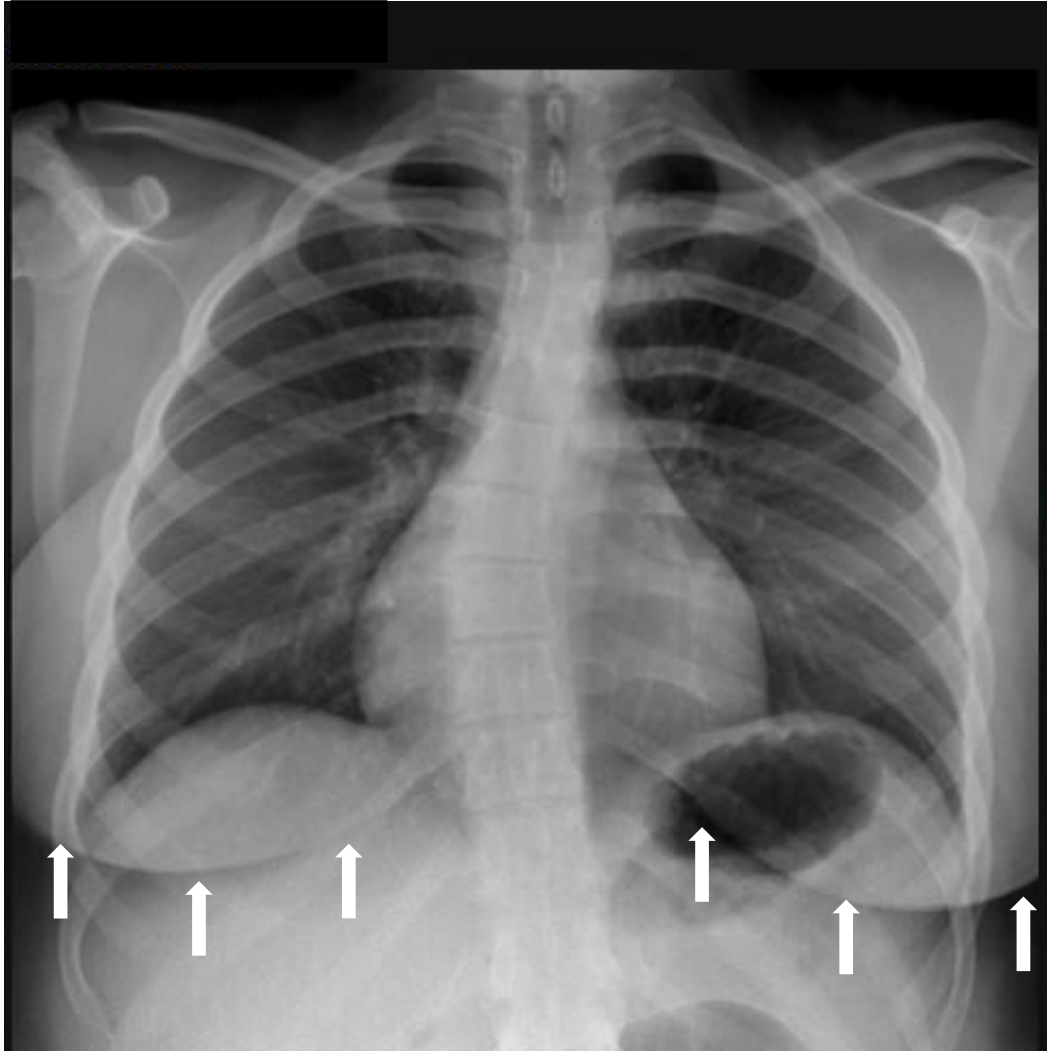


PCP and nipple ring

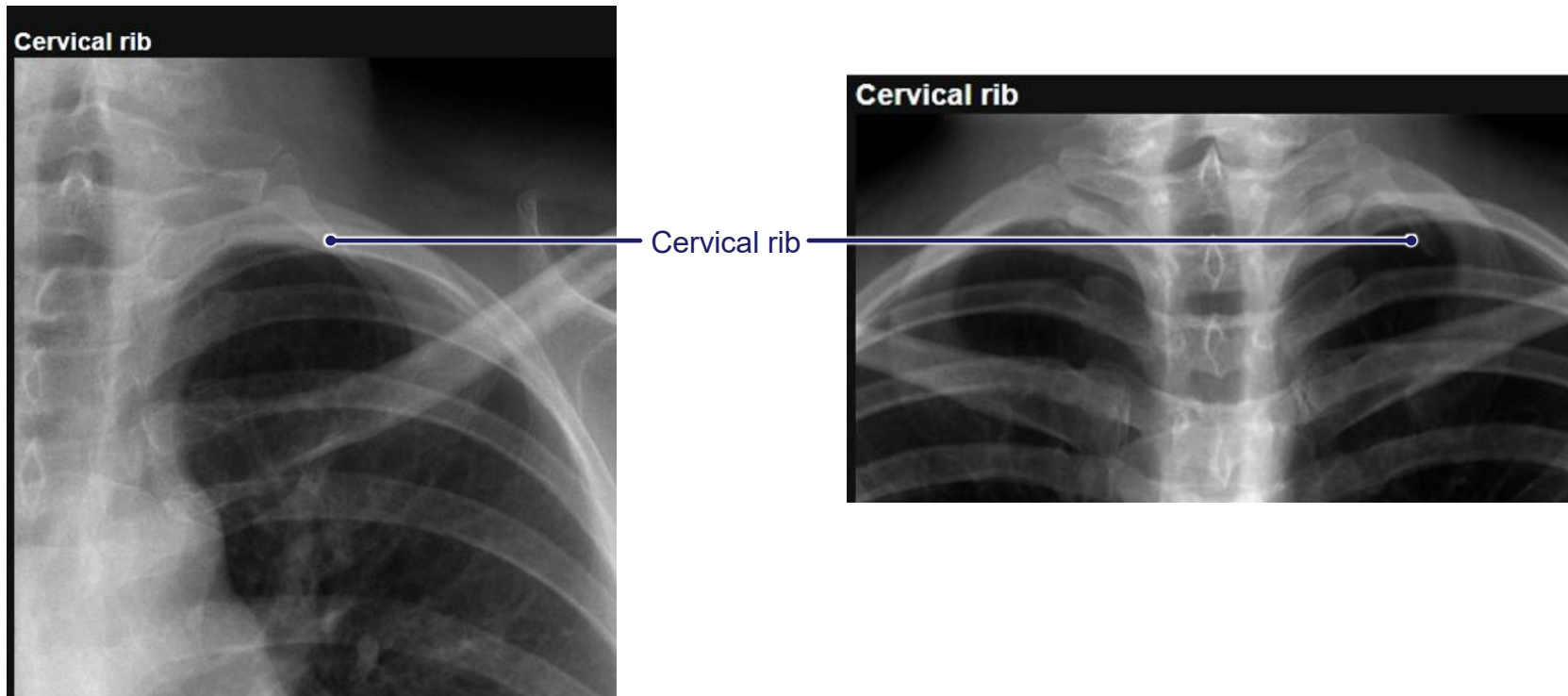


Chest X-ray

The mammary glands (white arrows) increase the density of structures posterior to them.



Anatomical Variation

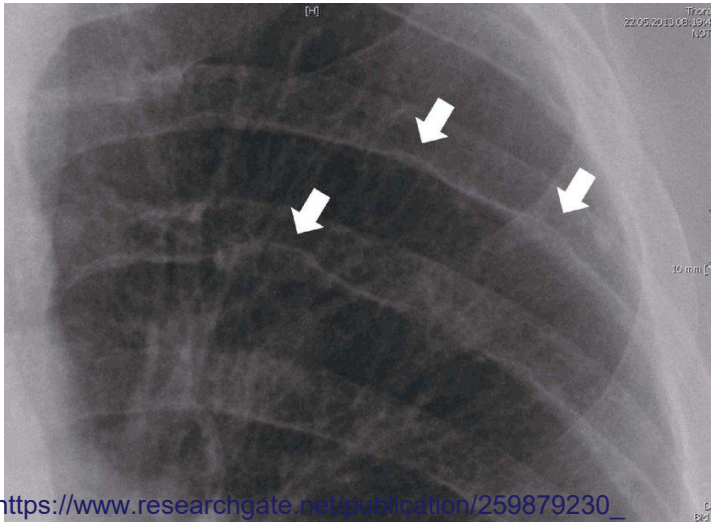


- A **cervical rib** is an extra rib that arises from the seventh cervical vertebra (C7). It's a congenital anomaly found in about 0.5–1% of the population.
- Normally, ribs attach to thoracic vertebrae, but in this case, an extra rib projects from C7.
- It may be small and fibrous, or it can extend down to fuse with the first rib.
- X-ray of the cervical spine or chest often shows the extra rib.

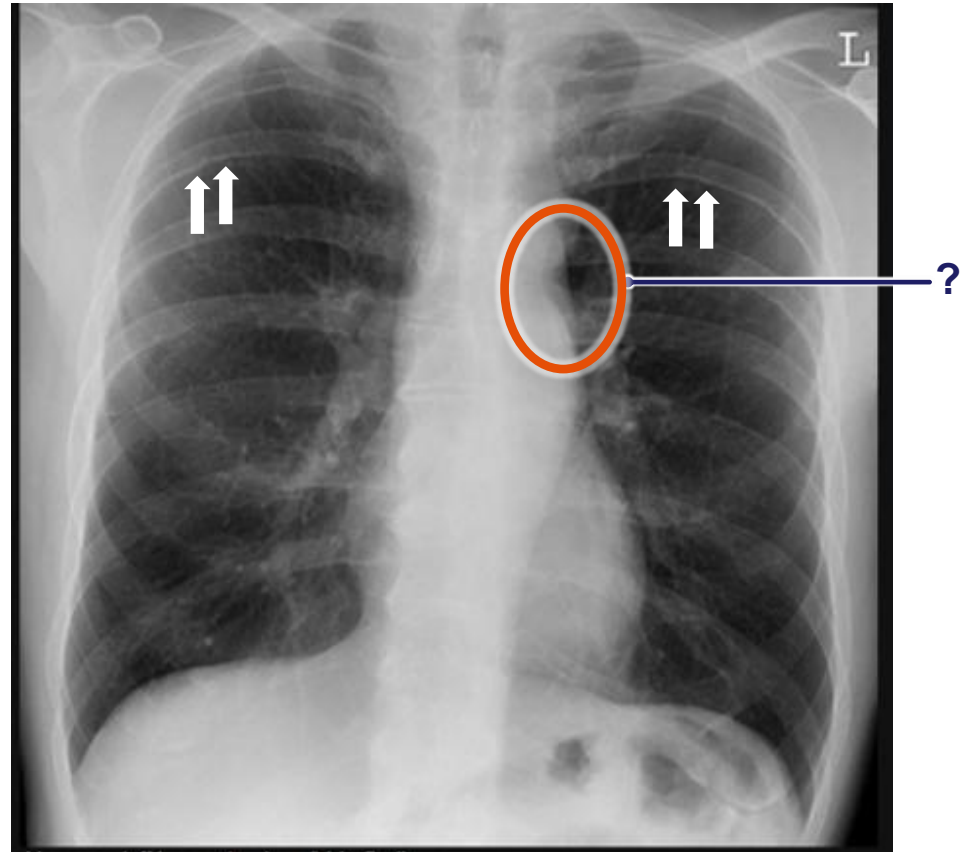
Notching of ribs

Rib notching = erosion or indentation seen along the inferior surface of the ribs on chest X-ray. It's usually due to **enlargement of intercostal arteries/veins** causing pressure on the rib.

- Usually caused by Coarctation of Aorta with complete aortic occlusion



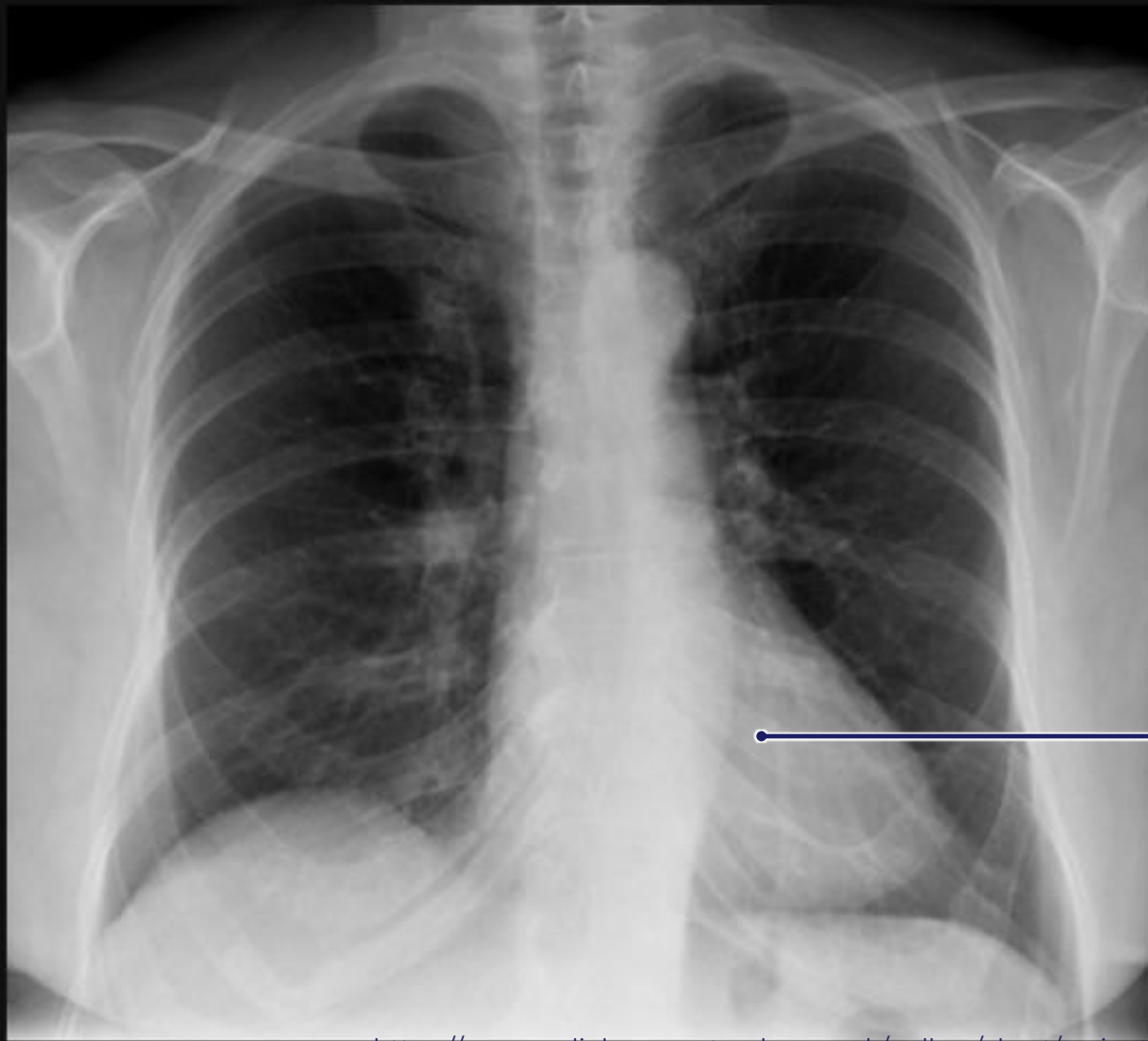
<https://www.researchgate.net/publication/259879230>
Coarctation_of_aorta_with_complete_aortic_occlusion



https://www.radiologymasterclass.co.uk/gallery/chest/mediastinum_hilum/coarctation#top_1st_img

Chest X-ray

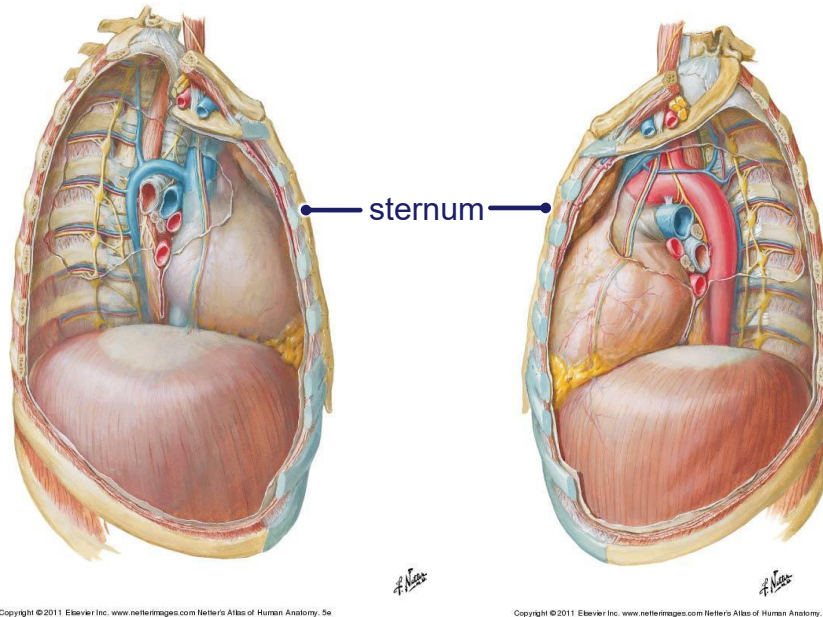
Costochondral calcification



Costochondral
calcification

https://www.radiologymasterclass.co.uk/gallery/chest/variants/costochondral_calcification

Lateral Chest View



A. Right lateral view

B. Left lateral view



C. Lateral x-ray

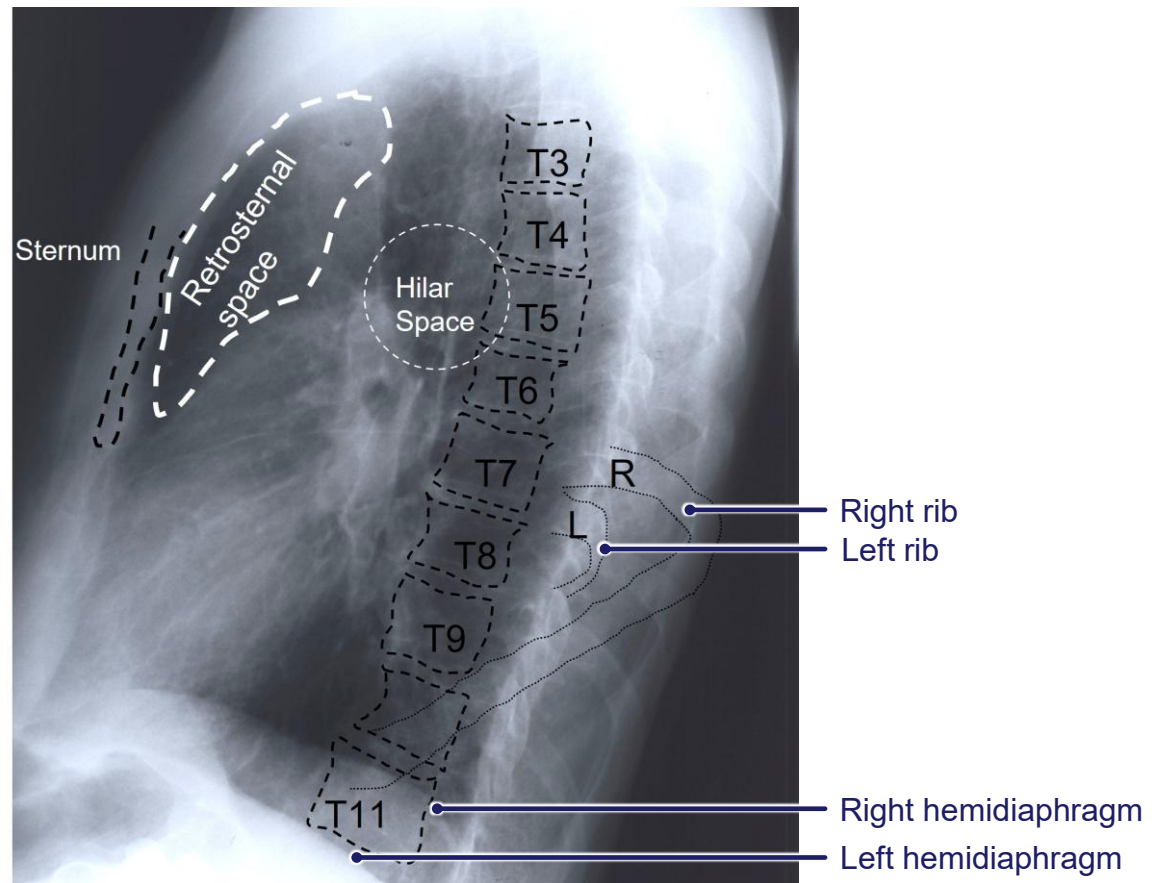
Lateral Thoracic X-ray

Assesses

- Retrosternal space
- Retrocardiac space
- Posterior costophrenic recesses

Features

- Spine appears to be darker caudally
- More air in lower lung lobes and thoracic wall is thinner
- Sternum should be seen edge on
- Posteriorly you should see two sets of ribs

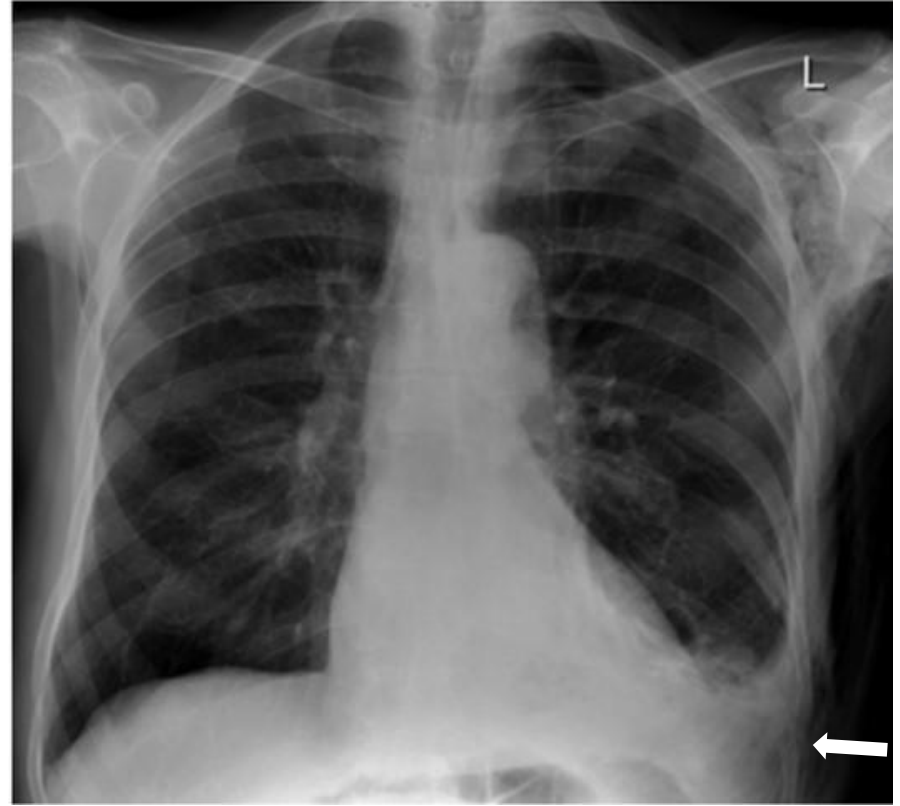
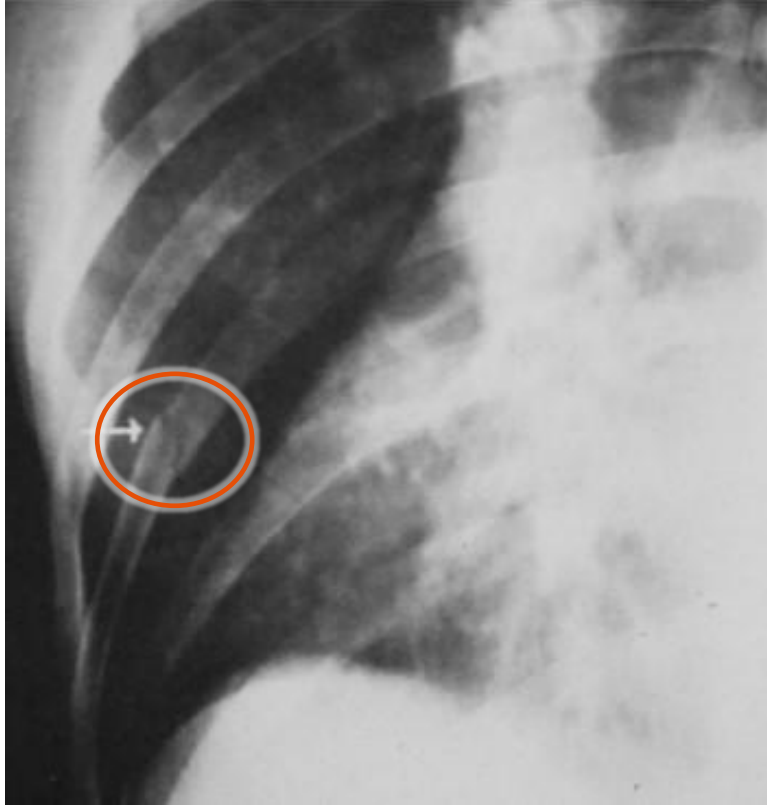


Chest X-ray- Hemidiaphragm, sternum



https://www.radiologymasterclass.co.uk/gallery/chest/variants/pectus_excavatum

Rib fractures

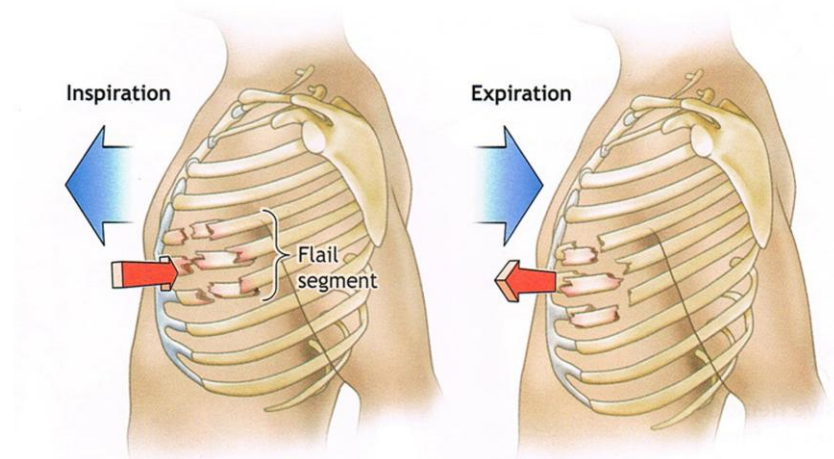
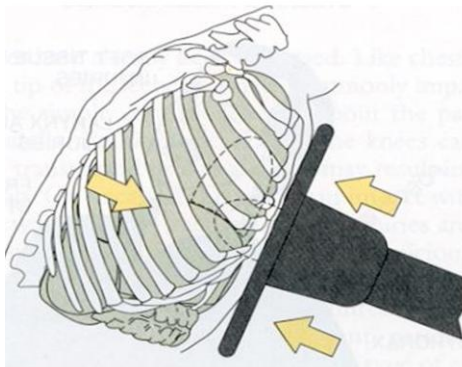


Multiple point fractures

- Associated with trauma; specifically vehicular accidents and pathology
- When pressure is applied from anterior, ribs may fracture in more than one place
- These fractures cause instability to the chest wall and may be associated with trauma to the pleura and/or blood vessels

Consequences

- Pneumothorax (air in the pleural cavity)
- Hemothorax (blood in the pleural cavity)



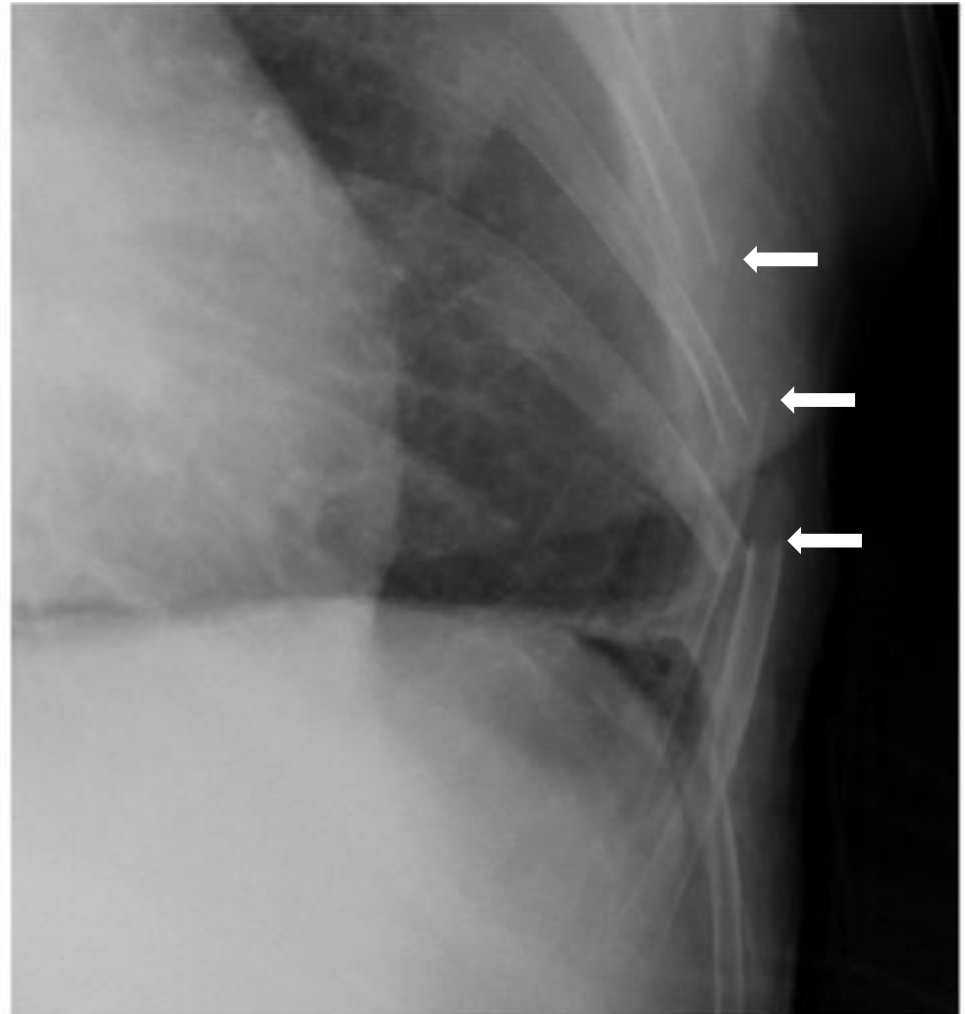
Flail Chest

Mechanism of action

- Multipoint fracture of consecutive ribs
- Segment becomes loose from rest of chest wall

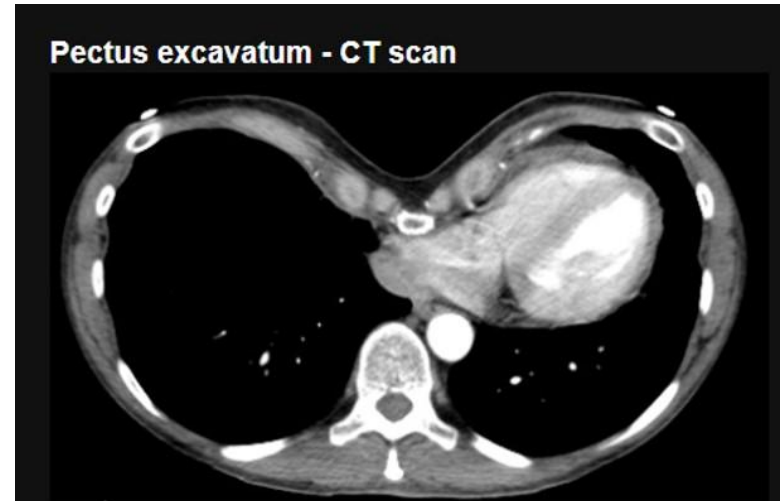
Clinical (radiographic) signs and symptoms

- Patient presents with shallow labored breathing
- The fractured segment(s) move in opposite direction from remainder of chest when patient breathes



CT Imaging of the Thorax- Axial view

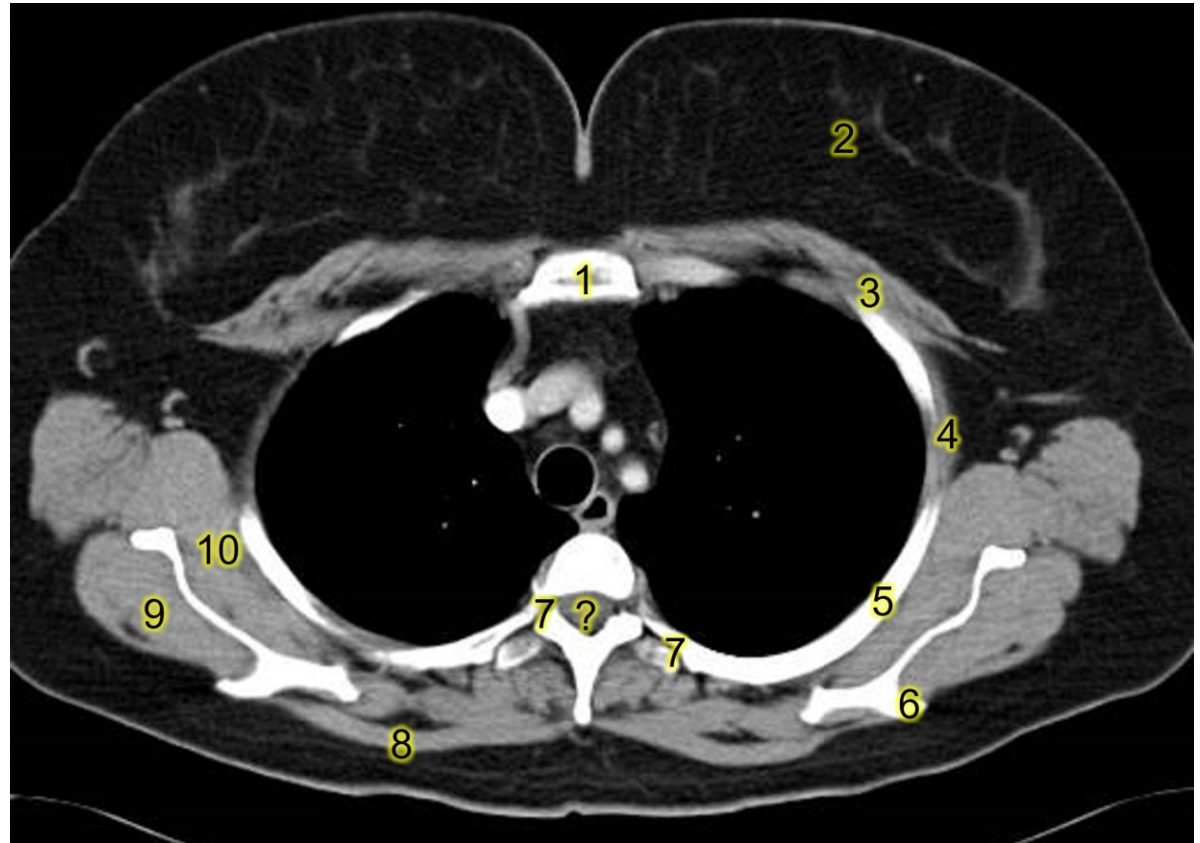
- Increased radiation
- Compiled of a series of slices
 - Coronal
 - Sagittal
 - Transverse (most common)
- Same rules apply as x-ray for what shows up except due to increased radiation soft tissues are easier to distinguish
- Clearer images
- Can be used for 3D reconstruction



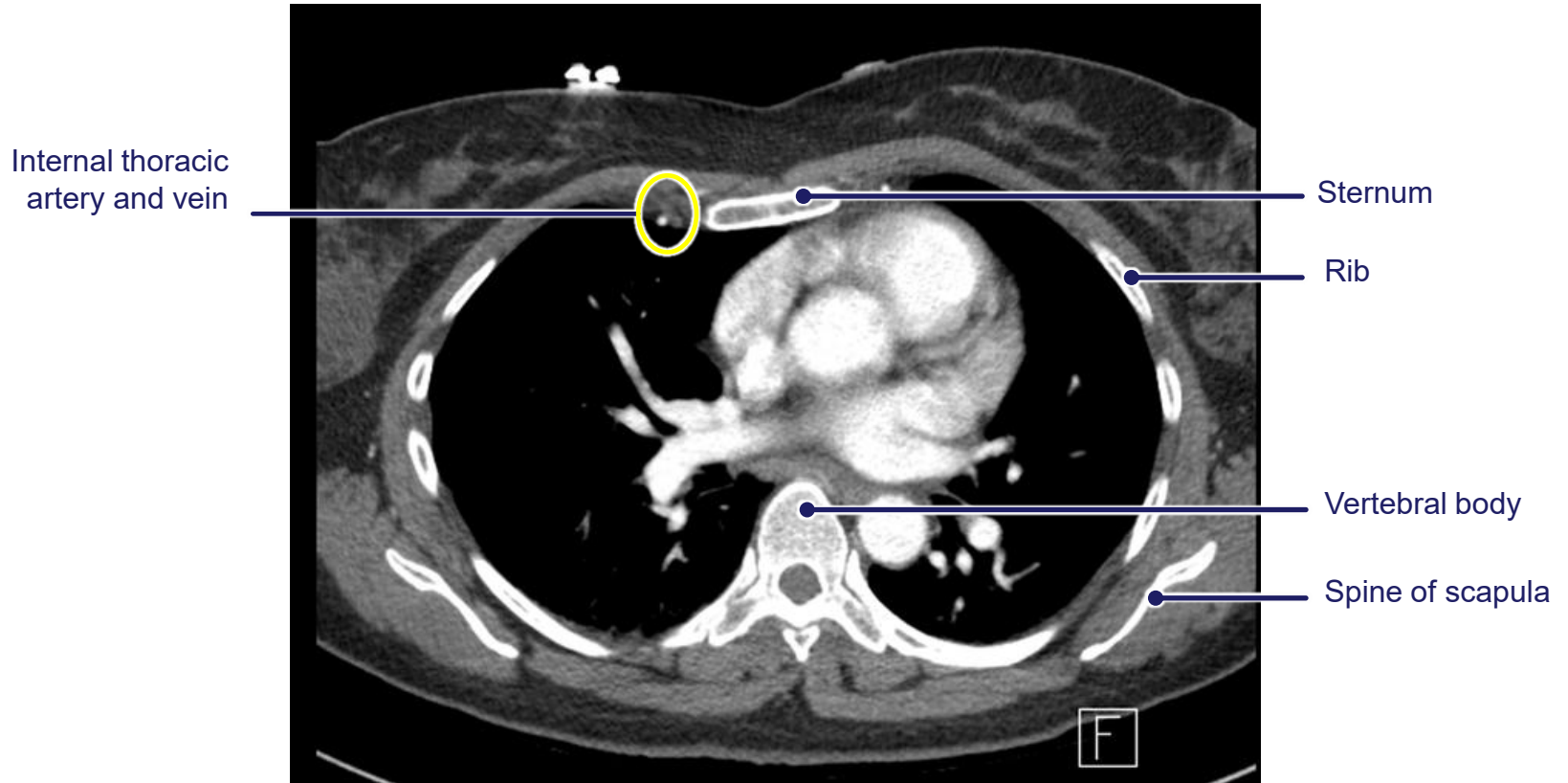
https://www.radiologymasterclass.co.uk/gallery/chest/variants/pectus_excavatum

Imaging of the Thorax- Axial view

- Breast
- Costovertebral joint
- Infraspinatus
- Pectoral muscles
- Rib
- Scapular spine
- Serratus anterior
- Sternum
- Subscapularis
- Trapezius

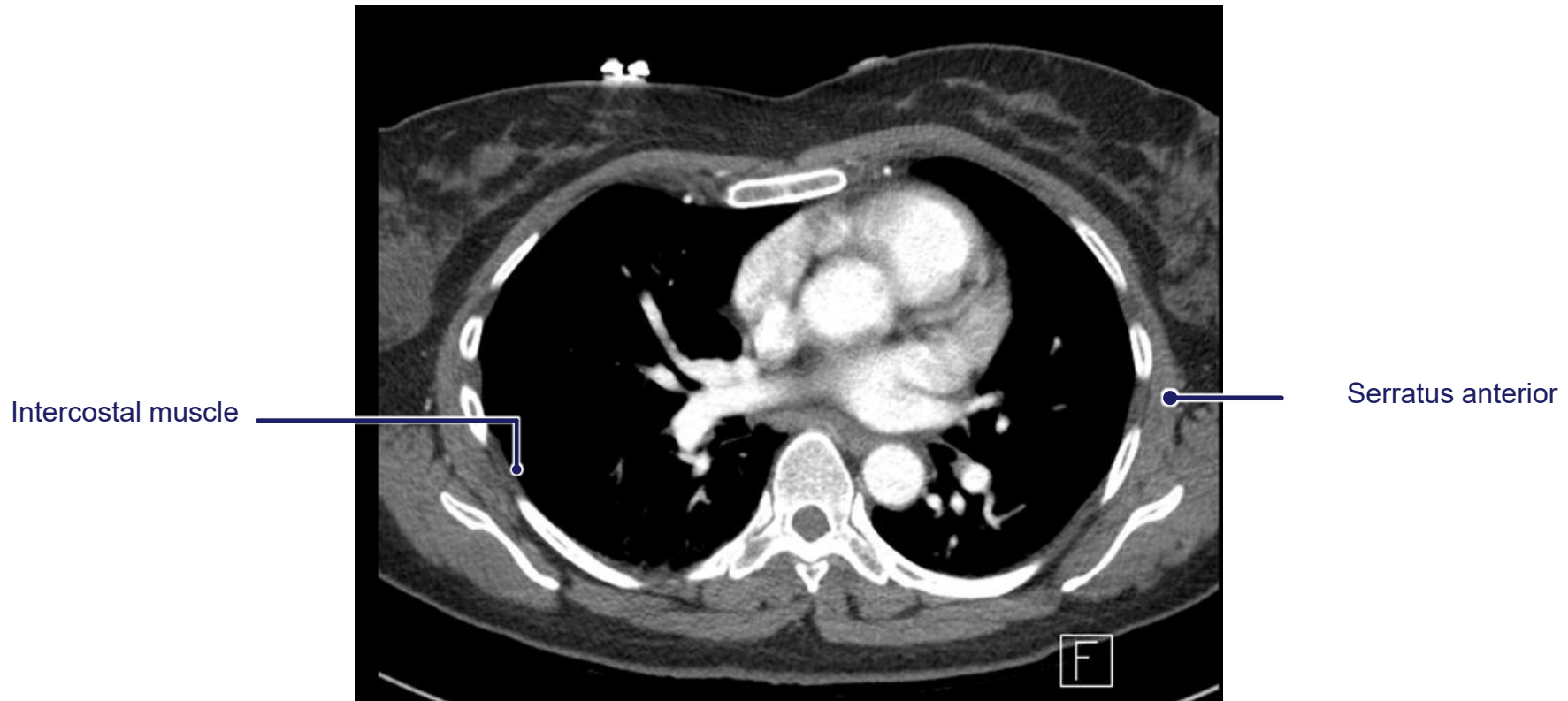


Imaging of the Thorax- Axial view



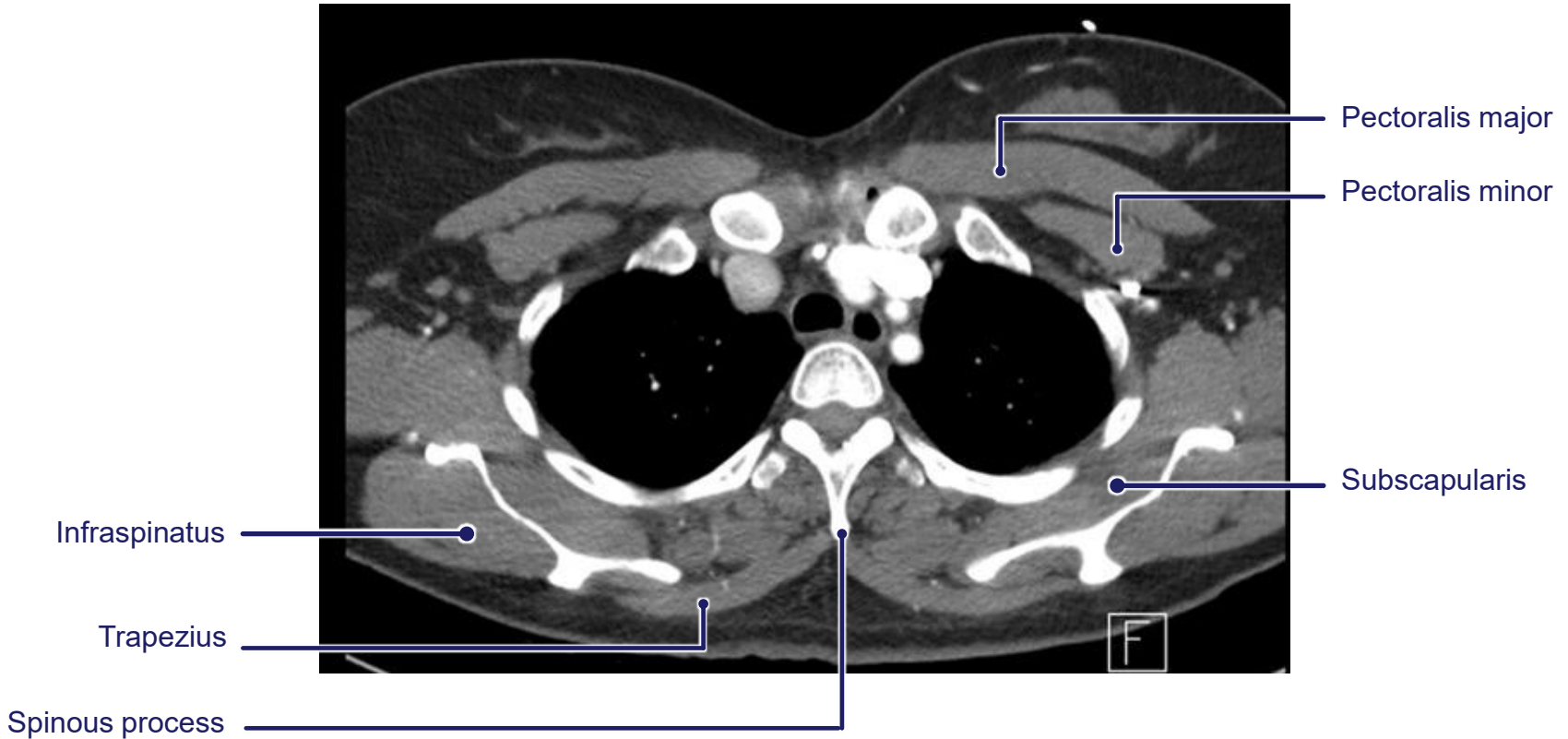
<https://radiopaedia.org/cases/62533/studies/70801?lang=us>

Imaging of the Thorax- Axial view



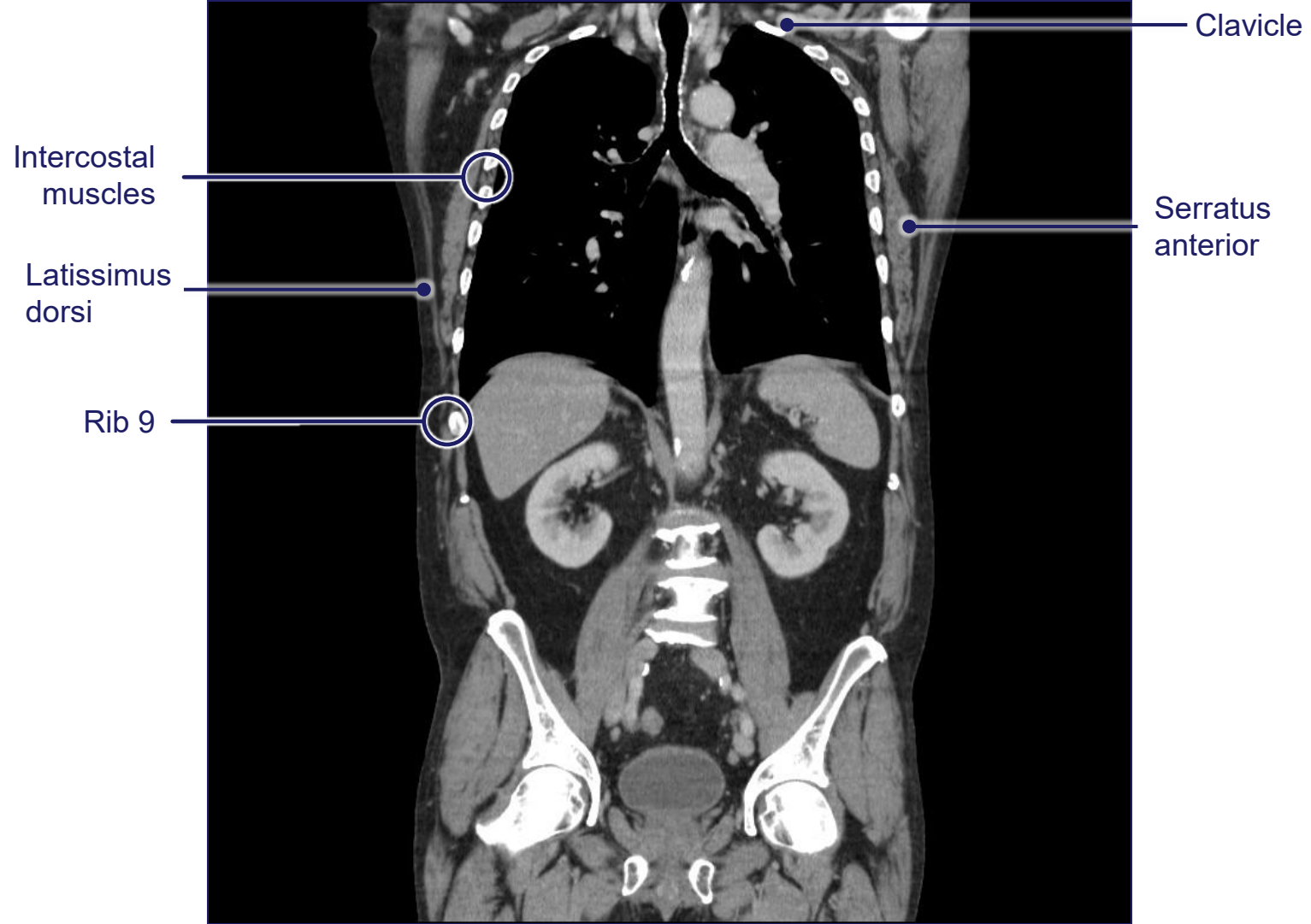
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Imaging of the Thorax- Axial view



<https://radiopaedia.org/cases/62533/studies/70801?lang=us>

Imaging of the Thorax- Coronal view



Imaging of the Thorax- Coronal view

